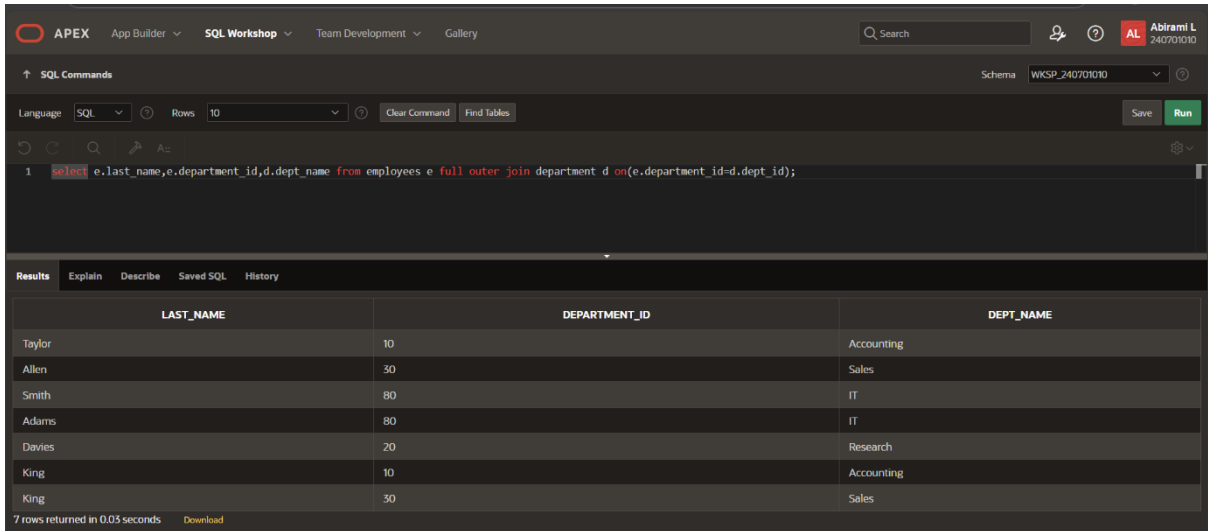


Exercise 7-Displaying data from multiple tables

1. Write a query to display the last name, department number, and department name for all employees.



The screenshot shows the APEX SQL Workshop interface. The SQL command entered is:

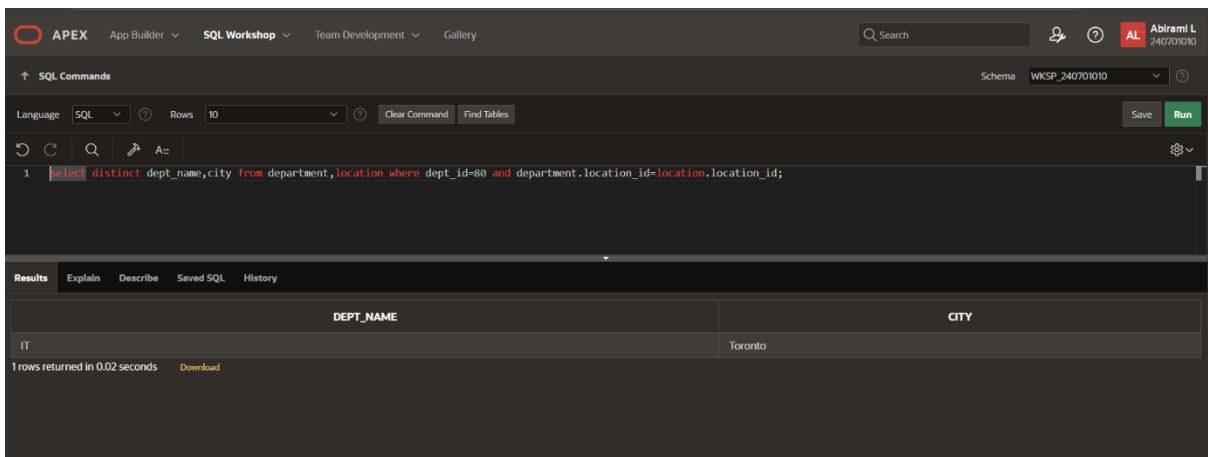
```
1 select e.last_name,e.department_id,d.dept_name from employees e full outer join department d on(e.department_id=d.dept_id);
```

The results are displayed in a table with the following data:

LAST_NAME	DEPARTMENT_ID	DEPT_NAME
Taylor	10	Accounting
Allen	30	Sales
Smith	80	IT
Adams	80	IT
Davies	20	Research
King	10	Accounting
King	30	Sales

7 rows returned in 0.03 seconds

2. Create a unique listing all of all jobs that are in department 80. Include the location of the department in the output.



The screenshot shows the APEX SQL Workshop interface. The SQL command entered is:

```
1 select distinct dept_name,city from department,location where dept_id=80 and department.location_id=location.location_id;
```

The results are displayed in a table with the following data:

DEPT_NAME	CITY
IT	Toronto

1 rows returned in 0.02 seconds

3. Write a query to display the employee last name, department name, location ID, and city of all employees who earn a commission

APEX App Builder SQL Workshop Team Development Gallery

Schema: WKSP_240701010

Language: SQL Rows: 10 Clear Command Find Tables Save Run

```
1 select last_name,department.dept_name,location.location_id,city from employees join department on employees.department_id=department.dept_id join location on department.location_id=location.location_id;
```

Results Explain Describe Saved SQL History

LAST_NAME	DEPT_NAME	LOCATION_ID	CITY
Adams	IT	4	Toronto
Allen	Sales	3	Toronto
King	Sales	3	Toronto
Davies	Research	2	Boston

4 rows returned in 0.03 seconds Download

4. Display the employee last name and department name for all employees who have an a(lowercase) in their last names. P

APEX App Builder SQL Workshop Team Development Gallery

Schema: WKSP_240701010

Language: SQL Rows: 10 Clear Command Find Tables Save Run

```
1 select last_name,dept_name from employees e,department d where e.department_id=d.dept_id and last_name like '%a%';
```

Results Explain Describe Saved SQL History

LAST_NAME	DEPT_NAME
Taylor	Accounting
Davies	Research
Adams	IT

3 rows returned in 0.01 seconds Download

5. Write a query to display the last name, job, department number, and department name for all employees who work in Toronto.

APEX App Builder SQL Workshop Team Development Gallery

Schema: WKSP_240701010

Language: SQL Rows: 10 Clear Command Find Tables Save Run

```
1 select last_name,job_id,department_id,dept_name from employees e join department d on e.department_id=d.dept_id join location l on d.location_id=l.location_id where city = 'Toronto';
```

Results Explain Describe Saved SQL History

LAST_NAME	JOB_ID	DEPARTMENT_ID	DEPT_NAME
Allen	SA_REP	30	Sales
Smith	IT_PROG	80	IT
Adams	IT_PROG	80	IT
King	IT_PROG	30	Sales

4 rows returned in 0.01 seconds Download

6. Display the employee last name and employee number along with their manager's last name and manager number. Label the columns as Employee, Emp#, Manager, Mgr# respectively

APEX App Builder SQL Workshop Team Development Gallery

Schema: WKSP_240701010

Language: SQL Rows: 10 Clear Command Find Tables Save Run

```
1 select last_name as Employee, employee_id as Emp#, manager_id as Mgr# from employees;
```

Results Explain Describe Saved SQL History

EMPLOYEE	EMP#	MGR#
Taylor	104	1001
Allen	103	1003
Smith	101	1004
Adams	102	1004
Davies	107	1003
King	1003	-
King	1004	-

7 rows returned in 0.00 seconds Download

7. To display all employees including King, who has no manager. Order the results by the employee number.

APEX App Builder SQL Workshop Team Development Gallery

Schema: WKSP_240701010

Language: SQL Rows: 10 Clear Command Find Tables Save Run

```
1 select last_name, dept_name from employees e, department d where e.manager_id is null and e.department_id = d.dept_id order by employee_id;
```

Results Explain Describe Saved SQL History

LAST_NAME	DEPT_NAME
King	Accounting
King	Sales

2 rows returned in 0.01 seconds Download

8. Create a query that displays employee last names, department numbers, and all the employees who work in the same department as a given employee. Give each column an appropriate label.

APEX App Builder SQL Workshop Team Development Gallery

Schema: WKSP_240701010

Language: SQL Rows: 10 Clear Command Find Tables Save Run

```
1 select e.last_name, d.dept_name from employees e join department d on e.department_id = d.dept_id where e.department_id = (select department_id from employees where last_name='Adams');
```

Results Explain Describe Saved SQL History

LAST_NAME	DEPT_NAME
Smith	IT
Adams	IT

2 rows returned in 0.02 seconds Download

9. Show the structure of the JOB_GRADES table. Create a query that displays the name, job, department name, salary, and grade for all the employees.

APEX SQL Workshop interface showing the 'Describe' tab for the 'JOB_GRADES' table.

SQL Command: `desc job_grades;`

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
JOB_GRADES	GRADE_LEVEL	VARCHAR2	2	-	-	-	✓	-	-
	LOWEST_SAL	NUMBER	22	-	-	-	✓	-	-
	HIGHEST_SAL	NUMBER	22	-	-	-	✓	-	-

APEX SQL Workshop interface showing the 'Results' tab for a query.

SQL Command: `select last_name, job_id, dept_name, salary, grade_level from employees e join department d on e.department_id = d.dept_id join job_grades j on salary between j.lowest_sal and j.highest_sal;`

LAST_NAME	JOB_ID	DEPT_NAME	SALARY	GRADE_LEVEL
Allen	SA_REP	Sales	5500	A
King	IT_PROG	Sales	8000	B
King	SALES_REP	Accounting	5500	A
Taylor	AC_ACCOUNT	Accounting	7000	B
Davies	FIN_MGR	Research	7500	B
Smith	IT_PROG	IT	6000	A
Adams	IT_PROG	IT	6500	B

7 rows returned in 0.02 seconds

10. Create a query to display the name and hire date of any employee hired after employee Davies.

APEX SQL Workshop interface showing the 'Results' tab for a query.

SQL Command: `select last_name as Employee_name, hire_date from employees where hire_date > (select hire_date from employees where last_name = 'Davies');`

EMPLOYEE_NAME	HIRE_DATE
Taylor	8/30/2017
Allen	3/20/2018
Smith	1/15/2020
Adams	7/10/2019
King	3/18/2019

5 rows returned in 0.01 seconds

11. Display the names and hire dates for all employees who were hired before their managers, along with their manager's names and hire date. Label the columns Employee, Emp hire, Manager, and Mgr hired respectively.

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SQL Commands

Schema: WKSP_240701010

Language: SQL Rows: 10

Clear Command Find Tables

Save Run

1 SELECT

2 e.last_name AS Employee,

3 e.hire_date AS Emp_hired,

4 m.last_name AS Manager,

5 m.hire_date AS Mgr_hired

6 FROM employees e

7 JOIN employees m

8 ON e.manager_id = m.employee_id

9 WHERE e.hire_date < m.hire_date;

10

Results Explain Describe Saved SQL History

EMPLOYEE	EMP_HIRED	MANAGER	MGR_HIRED
Allen	3/20/2018	King	3/18/2019
Davies	3/18/2017	King	3/18/2019

2 rows returned in 0.01 seconds Download